

An Inverse Distance Ladder from Public DESI BAO: H_0 from DESI DR2 BAO plus a BBN ω_b Prior, with an Exact Boltzmann-Code Sound Horizon

AutoevolveAI / ANSE Research Pipeline

Generated with Claude Code; model-produced. The model name recorded by
the generating session (Claude Opus 5.5) is an unverifiable statement:

no artifact can confirm which model produced earlier stages.

Not reviewed by a person

September 27, 2026

Abstract

We reproduce the DESI inverse-distance-ladder measurement of the Hubble constant. The inputs are the public DESI DR2 Gaussian BAO summary statistics [1] and a Big-Bang-nucleosynthesis prior $\omega_b = 0.02218 \pm 0.00055$. No CMB anisotropy or θ_* information enters. The fit assumes flat Λ CDM with standard pre-recombination physics.

The analysis was preregistered with a fitting-formula sound horizon. A protocol amendment then made an exact CAMB 2.0.4 r_d the primary analysis, cross-checked with CLASS. The preregistered fitting formula is kept as the secondary analysis.

The primary analysis gives $H_0 = 68.55 \pm 0.59 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\Omega_m = 0.2977 \pm 0.0085$. The published values are 68.51 ± 0.58 and 0.2977 ± 0.0086 . The preregistered verdict is **PASS**: $|\Delta H_0| = 0.035$, against a strict tolerance of 0.15.

The DESI DR1 second check is **PARTIAL**. It gives $H_0 = 68.70 \pm 0.81$ against the published 68.53 ± 0.80 . $|\Delta H_0| = 0.167$ exceeds the strict 0.15 but lies within the soft $0.33 \text{ km s}^{-1} \text{ Mpc}^{-1}$. It crosses the strict limit by 0.017, which is about 1.1 times our Monte-Carlo error on the mean combined with the rounding of the published value, so the PASS/PARTIAL boundary is not statistically resolved.

The secondary analysis gives the same verdicts: T1 PASS with $H_0 = 68.58 \pm 0.60$, and T2 PARTIAL. It is bit-identical to a hash-locked fit that was written before the amendment.

Four positive controls and three negative controls behave as preregistered (P2 with one clarification declared before it ran). N1 and N2 would fail for almost any pipeline; a post-hoc control added after review (N4, a neutrino-bookkeeping error in r_d) shifts H_0 by $+0.164$ and is invisible to the data ($|\Delta\chi^2| = 5.2 \times 10^{-5}$). The BAO-only gate reproduces DESI's (Ω_m, hr_d) with pulls 0.024σ and -0.027σ . 10 Lean 4 theorems about the model as coded are kernel-checked with only the trusted axioms. Six are substantive: positivity of the eq. 16 sound horizon, three strict-monotonicity results on the sampled support, and the preregistered identifiability statement that $hr_d(h)$ is strictly increasing in h (with its generic lemma). The other four are the positivity of a constant and three field identities.

This is a reproduction of a published number from its own public inputs. It is not a new constraint.

1 Introduction

BAO measure $D_M(z)/r_d$, $D_H(z)/r_d = c/[H(z)r_d]$ and $D_V(z)/r_d$. In flat Λ CDM these fix Ω_m and the product hr_d , but not h and r_d separately. With standard early-universe physics, r_d is a function of (ω_b, ω_{cb}) . A BBN prior on ω_b therefore calibrates the ruler and yields H_0 without the CMB acoustic scale [1, 5]. DESI DR2 reports $H_0 = 68.51 \pm 0.58$ (eq. 19 and Table V) and DESI DR1 reports 68.53 ± 0.80 [2]. We re-derive both from the released summary statistics and treat agreement as an external correctness test of the pipeline. We then state what the agreement does and does not show.

2 Data

We use `desi_gaussian_bao_ALL_GCcomb_{mean,cov}.txt` from DESI DR2: 13 points and a full 13×13 covariance (Table 1). For DR1 we use `desi_2024_gaussian_bao_ALL_GCcomb_{mean,cov}.txt`: 12 points, DESI only, no SDSS [2]. The sha256 of all four files was fixed in the preregistration before any fit and re-verified by the fit script. They are recorded in `results/bao_bbn_h0/fit.json`.

Table 1: DESI DR2 BAO data (values and $\sqrt{\text{diag } C}$ read from the data files).

z_{eff}	Quantity	Value	σ
0.295	D_V/r_d	7.942	0.076
0.510	D_M/r_d	13.588	0.168
0.510	D_H/r_d	21.863	0.429
0.706	D_M/r_d	17.351	0.180
0.706	D_H/r_d	19.455	0.334
0.934	D_M/r_d	21.576	0.162
0.934	D_H/r_d	17.641	0.201
1.321	D_M/r_d	27.601	0.325
1.321	D_H/r_d	14.176	0.225
1.484	D_M/r_d	30.512	0.764
1.484	D_H/r_d	12.817	0.518
2.330	D_H/r_d	8.632	0.101
2.330	D_M/r_d	38.989	0.532

3 Model and method

Parameters and priors. We sample $(H_0, \omega_b, \Omega_m)$ with emcee (32 walkers). The flat priors are $H_0 \in [20, 100]$, $\omega_b \in [0.005, 0.1]$ and $\Omega_m \in [0.01, 0.99]$, as in DESI DR1 Table 2 [2], which DR2 adopts [1]. On top of these sits the Gaussian BBN prior $\omega_b = 0.02218 \pm 0.00055$ [1, 4]. We fix $N_{\text{eff}} = 3.044$, one massive neutrino of 0.06 eV, and $T_{\text{CMB}} = 2.7255$ K. Ω_m includes the massive neutrino.

We run until $N_{\text{steps}} > 50\tau$ for every parameter and discard 5τ as burn-in. The reported statistic is the posterior mean and standard deviation, as in DR2 Table V. We also report the MAP, a Laplace width and a profile likelihood.

Table 2: DESI DR1 BAO data used for the second check (read from the data files).

z_{eff}	Quantity	Value	σ
0.295	D_V/r_d	7.925	0.151
0.510	D_M/r_d	13.620	0.252
0.510	D_H/r_d	20.983	0.611
0.706	D_M/r_d	16.846	0.319
0.706	D_H/r_d	20.079	0.595
0.930	D_M/r_d	21.708	0.282
0.930	D_H/r_d	17.876	0.346
1.317	D_M/r_d	27.787	0.690
1.317	D_H/r_d	13.824	0.422
1.491	D_V/r_d	26.072	0.669
2.330	D_M/r_d	39.708	0.943
2.330	D_H/r_d	8.523	0.171

Background. D_M and D_H come from a Gauss–Legendre integrator of $E(z) = [\Omega_m(1+z)^3 + \Omega_r(1+z)^4 + 1 - \Omega_m - \Omega_r]^{1/2}$. Here Ω_r holds the photons and the massless share of N_{eff} , and the 0.06 eV neutrino is counted as matter. This background agrees with astropy FlatLambdaCDM to 1.7×10^{-6} (control P4). It agrees with CAMB’s own background D_M and D_H to 1.2×10^{-6} .

Primary sound horizon (amendment 1). r_{drag} is computed with CAMB 2.0.4 at $N_{\text{eff}} = 3.044$ with one massive neutrino, on a $49 \times 61 \times 3$ grid in $(\ln \omega_b, \ln \omega_{\text{cdm}}, h)$ (8967 nodes). The interpolant is a bicubic spline in $\ln r_{\text{drag}}$ at each h node, linear in h . It matches direct CAMB calls to a maximum of 3.1×10^{-5} (rms 2.3×10^{-6}) at 200 core points, and to 6.4×10^{-6} at 100 wide-range points. CLASS 3.3.4 agrees with CAMB to at most 7.0×10^{-5} at five points, which include the BBN $\pm 2\sigma$ edges.

22 nodes failed in CAMB’s recombination solver. All of them sit at the $h = 0.99$ node with very low Ω_m . They were refilled with CAMB at $h = 0.98$, and every refilled node is listed in `rd_camb_validation.json`.

At fixed physical densities, the table’s r_{drag} varies across its h nodes by at most 1.8×10^{-5} in the core region. The largest variation, 2.1×10^{-4} , is at the $h = 0.2$ node in a far corner of the grid, not at a refilled node.

At the Planck 2018 Table 1 best-fit inputs ($N_{\text{eff}} = 3.046$) the instrument gives $r_{\text{drag}} = 147.0492$ Mpc. That reproduces the published 147.049 Mpc [6].

Secondary sound horizon (preregistered). The secondary analysis uses Aubourg et al. eq. 16 [3]:

$$r_d \simeq \frac{55.154 e^{-72.3(\omega_\nu + 0.0006)^2}}{\omega_{cb}^{0.25351} \omega_b^{0.12807}} \text{ Mpc}, \quad \omega_\nu = 0.0107 \times 0.06, \quad \omega_{cb} = \Omega_m h^2 - \omega_\nu. \quad (1)$$

Its stated accuracy is 0.021% for $N_{\text{eff}} = 3.046$ within 3σ of Planck [3]. We measured eq. (1) against CAMB at $N_{\text{eff}} = 3.044$ at the five cross-check points. The difference ranges from -0.029% to $+0.006\%$. Its magnitude reaches 0.029%, which is above the stated 0.021%. About 0.0065% of that is the uncorrected N_{eff} offset from Aubourg’s eq. 17. The rest is unexplained here.

BAO-only gate. Gates G1 (DR2) and G2 (DR1) sample (Ω_m, hr_d) with no r_d model and $hr_d \in [10, 1000]$. The radiation term is evaluated at a fixed $h = 0.6851$. These gates are the external positive control against DESI’s own pipeline.

4 Preregistration, amendment and locked output

The preregistration (`preregistration.json`, sha256 5bcde156f38c2218..., 2026-09-27 15:53:42 UTC) fixed these choices before any fit: targets, priors, statistic, r_d method, controls and tolerances.

Tolerances. DESI used the same data vector, BBN prior and flat priors. The published σ is therefore not independent noise on the difference, so the tolerances were built from a systematic budget instead:

- **Strict:** $0.15 \text{ km s}^{-1} \text{ Mpc}^{-1}$. This is a 0.10% r_d -formula proxy (0.139) added in quadrature to a 0.05 Monte-Carlo allowance, i.e. 0.26σ of T1.
- **Soft:** $0.33 \text{ km s}^{-1} \text{ Mpc}^{-1}$, from the maximum 0.233% difference between two r_d formulas over the prior box.

Amendment 1 kept these tolerances unchanged, but the 0.139 formula term no longer applies to the primary analysis, which uses CAMB. With only the preregistered Monte-Carlo allowance (0.05) and the measured table interpolation error propagated to H_0 (0.004), the strict budget for the primary would be $0.050 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The preregistered strict limit of 0.15 is therefore generous for the primary. We apply the preregistered rule unchanged. For information, under the re-derived 0.050 T1 passes and T2 fails (Sec. 6).

Verdict rule. PASS requires all of the following:

- $|H_0 - 68.51| \leq 0.15$;
- $|\Omega_m - 0.2977| \leq 0.25 \times 0.0086$;
- $\sigma(H_0)$ within $\pm 15\%$ of 0.58;
- gate G1 passes;
- every control behaves.

PARTIAL means $0.15 < |\Delta H_0| \leq 0.33$ with the other conditions met. The headline verdict is T1. T2 (DR1) is reported separately under the same thresholds.

Timeline of the amendment. The events, taken from the amendment file:

- CAMB, CLASS and cobaya were installed after the preregistration was written.
- A fitting-formula fit output was written at 16:03:03 UTC.
- Amendment 1 was filed at 2026-09-27 16:04:27 UTC. It made the exact CAMB r_d primary and kept the fitting formula as the secondary analysis. Targets, tolerances and controls were unchanged.

- The amender states that the earlier output was not read. This is an assertion that no hash can prove; we label it unverifiable. Both analyses give the same verdicts, so no selection between them was possible. The output was hash-locked and renamed (`fit_attempt1_fittingformula_UNREAD_AT_AMENDMENT.json`, sha256 `b7fe5608e4ae6d0b...`).

The locked output. It contains T1 $H_0 = 68.579 \pm 0.601$ and T2 $H_0 = 68.701 \pm 0.815$. Our re-run secondary analysis reproduces it bit for bit: the differences are 0.0 (T1) and 0.0 (T2), and G1 and G2 are identical. The seeds and code path are fixed, so this shows reproducibility, not independent correctness.

The amendment’s reason quotes a CAMB positive control of 147.1027 Mpc, a CLASS value of 147.0971 Mpc, and a published Planck 2018 value of 147.09 ± 0.26 . That published value is the posterior mean in [6] Table 1 (column “Plik [1]”, the same row as the 147.049 best fit) and Table 2 (TT,TE,EE+lowE+lensing); [2] quotes it before its eq. (4.3). Both code values are reproduced at the Table 2 TT,TE,EE+lowE+lensing means ($H_0 = 67.36$, $\omega_b = 0.02237$, $\omega_c = 0.1200$, one 0.06 eV neutrino). CAMB with $N_{\text{eff}} = 3.044$ gives 147.1027 Mpc. CLASS with $N_{\text{ur}} = 2.0328$ gives 147.0971 Mpc, but CLASS reports $N_{\text{eff}} = 3.046$ for that setting. So the amendment’s two numbers are not at the same N_{eff} . At matched $N_{\text{eff}} = 3.044$ ($N_{\text{ur}} = 2.0308$, the setting used for our CLASS cross-check) CLASS gives 147.1071 Mpc, a fractional difference of 3.0×10^{-5} from CAMB. The ledger stage re-ran these CAMB and CLASS evaluations; they are bit-identical to the rows recorded in `fit.json`. An earlier draft of this paper said the anchor could not be reproduced and that 147.09 was not in the fetched sources. Both statements were wrong and were corrected.

5 Controls

Each control was run for both analyses before the target comparison.

- **P1, noiseless mock** (truth $H_0 = 68$, $\Omega_m = 0.300$, $\omega_b = 0.02218$; rule $|\Delta H_0| \leq 0.1$, $|\Delta \Omega_m| \leq 0.002$). Primary: $\Delta H_0 = +0.0041$, $\Delta \Omega_m = +0.0006$. Secondary: $\Delta H_0 = +0.0035$, $\Delta \Omega_m = +0.0005$. Passed.
- **P2, 100 noisy mocks, MAP per mock** (rule: mean H_0 pull within ± 0.2 , width in $[0.8, 1.2]$). Primary: mean -0.064 , width 1.023. Secondary: mean -0.067 , width 1.025. The primary Ω_m pull is $-0.131/0.944$ and the mean χ^2 is 10.95 for 11 dof. Passed. *Deviation, written before P2 first ran:* each mock also draws the BBN centre from $\mathcal{N}(0.02218, 0.00055)$. Without that draw the pull width is below 1 by construction.
- **P3 = gate G1:** passed (Table 3). G2 also passed.
- **P4, predictor cross-check:** astropy against the manual integrator, maximum fractional difference 1.7×10^{-6} (threshold 10^{-4}). Passed.
- **N1, scrambled DR2 vector** (fixed seed): $\chi^2 = 1.43 \times 10^5$ (primary) and 1.35×10^5 (secondary), PTE 0.0. The real data give PTE 0.505. Behaves.
- **N2, Einstein–de Sitter** on the real data: $\chi^2 = 1420.2$ for 12 dof, PTE 6.2×10^{-297} . Behaves.
- **N3, no BBN prior** (rule: $\sigma(H_0)$ grows more than $5\times$): $\sigma(H_0)$ grows $20.99\times$ (primary) and $21.71\times$ (secondary). Behaves.

- The primary chain ran 40000 steps with $\tau_{\max} \approx 69$ ($N/\tau \approx 582$).
- The secondary chain ran 88000 steps with $\tau_{\max} \approx 1300$ ($N/\tau \approx 68$). A τ estimate for such a broad, prior-truncated posterior is not reliable, so the $50\text{-}\tau$ criterion is nominal here. The control’s conclusion does not depend on it: the ratio is far above 5.
- The posterior fills the flat priors. The CAMB table’s support ($h \leq 0.99$, $\omega_{\text{cdm}} \geq 0.005$) truncates the primary prior, and this matters only for N3.

The preregistration’s report rules mention N1–N4, but only N1–N3 are defined. The $N_{\text{eff}} = 4.044$ run is a sensitivity demonstration, not a control.

What the controls do and do not show. P1 and P2 use the same predictor that generated the mocks, so they test self-consistency, not correctness. N1 ($\chi^2 \sim 10^5$) and N2 would fail for almost any pipeline, including one with a wrong r_d convention or a wrong neutrino treatment. Of the preregistered controls, only N3 is informative, and its ratio depends on the prior box, which the CAMB table truncates for the primary. The external checks are G1/G2, the T1/T2 comparisons themselves, and the CLASS-versus-CAMB r_d cross-check. “All controls behave” should not be read as more than that.

N4, post-hoc wrong-convention control (not preregistered). N4 was added after review. It is not part of the verdict’s controls condition. Its expectation and pass rule were written to `controls/N4_expectation.json` before it ran. N4 evaluates the primary CAMB r_d with $\omega_{\text{cdm}} = \Omega_m h^2 - \omega_b$, forgetting to subtract the massive-neutrino density. On the DR2 BAO+BBN likelihood it shifts the MAP H_0 by $+0.164 \text{ km s}^{-1} \text{ Mpc}^{-1}$, which exceeds the strict 0.15. The fit quality is unchanged ($|\Delta\chi^2| = 5.2 \times 10^{-5}$). So the data cannot detect this error; only the external comparison can, and only by a narrow margin over the strict limit. The soft tolerance would not catch it.

6 Results

Table 3: Posterior means $\pm$ standard deviations (emcee), against the published values. $|\Delta H_0|$ is in $\text{km s}^{-1} \text{ Mpc}^{-1}$. All entries are generated from `fit.json`.

Target	Analysis	H_0	Ω_m	$ \Delta H_0 $	Verdict
T1 DR2+BBN	primary (CAMB)	68.55 ± 0.59	0.2977 ± 0.0085	0.035	PASS
	secondary (eq. 16)	68.58 ± 0.60	0.2976 ± 0.0086	0.069	PASS
	DESI DR2 [1]	68.51 ± 0.58	0.2977 ± 0.0086	–	–
T2 DR1+BBN	primary (CAMB)	68.70 ± 0.81	0.2953 ± 0.0146	0.167	PARTIAL
	secondary (eq. 16)	68.70 ± 0.81	0.2946 ± 0.0143	0.171	PARTIAL
	DESI DR1 [2]	68.53 ± 0.80	0.295 ± 0.015	–	–
		hr_d [Mpc]	Ω_m	pulls	
G1 DR2 BAO-only	shared	101.52 ± 0.74	0.2977 ± 0.0087	$-0.027, 0.024$	passed
	DESI DR2	101.54 ± 0.73	0.2975 ± 0.0086		
G2 DR1 BAO-only	shared	101.93 ± 1.27	0.2945 ± 0.0145	$0.103, -0.033$	passed
	DESI DR1	101.8 ± 1.3	0.295 ± 0.015		

Headline (primary, T1). $H_0 = 68.545 \pm 0.594$, which gives $|\Delta H_0| = 0.035 \leq 0.15$. The Ω_m pull is -0.0005σ . The ratio $\sigma(H_0)/0.58$ is 1.024. G1 passed and every control behaved. The verdict is **PASS**.

Two other estimators agree with the posterior mean:

- MAP $H_0 = 68.53$ with Laplace $\sigma = 0.60$;
- profile likelihood, 1σ interval $[67.93, 69.13]$.

$\chi^2_{\text{MAP}} = 10.28$ for 11 dof (PTE 0.51). The derived r_d is 148.1 ± 1.6 Mpc, and $\text{corr}(H_0, \omega_b) = 0.74$. The chains ran 4000 steps with $\tau_{\text{max}} \approx 39$ and ESS 3109.

Second check (T2, DR1). The primary gives $H_0 = 68.697 \pm 0.809$ and the secondary 68.701 ± 0.815 . $|\Delta H_0| = 0.167$ (primary) and 0.171 (secondary) exceed the strict 0.15 and lie within the soft 0.33. The Ω_m and σ conditions pass. Both analyses give **PARTIAL**.

The strict limit is crossed by 0.017 (primary) and 0.021 (secondary). Our Monte-Carlo error on the posterior mean, $\sigma/\sqrt{\text{ESS}}$, is 0.014 (primary) and 0.014 (secondary). The published 68.53 is rounded to 0.01, a half-unit of 0.005. The margin is therefore 1.1 (primary) and 1.4 (secondary) times this combined resolution. DESI's own Monte-Carlo error is not published and is not included. PASS versus PARTIAL for T2 is not statistically resolved. PARTIAL is what the preregistered rule gives, and it is what we report.

G2 passes its gate against DR1's BAO-only $hr_d = 101.8$, with an hr_d pull of 0.103. DR1 itself also quotes 101.9 ± 1.3 in its Sec. 6. A one-line propagation could account for the offset. At fixed (Ω_m, ω_b) we measure $d\ln(hr_d)/d\ln h = 0.492$ at the T2 primary mean. Our G2 value is $hr_d = 101.93$. If DESI's DR1 chain effectively sat at the Sec. 8 value 101.8, our H_0 would be high by $0.18 \text{ km s}^{-1} \text{ Mpc}^{-1}$, which is about the whole T2 offset. At the Sec. 6 value 101.9 it would be high by only 0.05. The PARTIAL is therefore plausibly a DESI-side Monte-Carlo or rounding difference in hr_d . That is consistent with the boundary not being statistically resolved. It is a heuristic, not a resolution. The preregistration labels this PARTIAL, and it stays PARTIAL.

Formula systematic, measured. Primary minus secondary gives $\Delta H_0 = -0.034$ (T1) and -0.004 (T2), well inside the preregistered 0.10% proxy (0.139). The sensitivity runs, each relative to the T1 mean of the analysis named:

- DESI eq. 2 power law in place of eq. 16: $+0.049$;
- no radiation in $H(z)$: -0.003 (secondary) and $+0.031$ (primary);
- $N_{\text{eff}} = 4.044$ (demonstration only): $+4.65$.

7 Lean 4 formalization

`formal/ANSE/BAO_BBN_H0.lean` (sha256 2b6530838efa0df9...) states exact facts about the *model as coded* in `scripts/bao_bbn_h0/common.py`. It uses pinned Mathlib imports (`Analysis.Real.Sqrt`, `SpecialFunctions.Pow.Real`, `Convex.SpecificFunctions.Basic`, `Tactic.{Positivity, Linarith, Ring}`) and no `import Mathlib`. The definitions, verbatim:

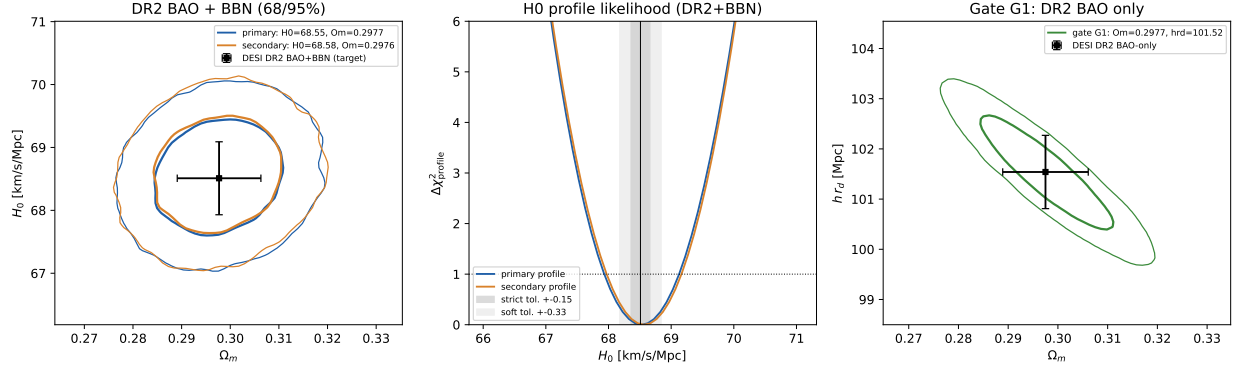


Figure 1: Left: DR2 BAO+BBN posterior (68/95%) for both analyses, with the DESI DR2 DESI+BBN target. Centre: H_0 profile likelihood, with the strict and soft tolerance bands shaded. Right: the G1 BAO-only gate in (Ω_m, hr_d) against DESI DR2 eq. 17.

```

noncomputable def omegaNu : ℝ := 0.0107 * 0.06

noncomputable def rdAubourg (wcb wb : ℝ) : ℝ :=
  55.154 * Real.exp (-72.3 * (omegaNu + 0.0006) ^ 2) /
    (wcb ^ (0.25351 : ℝ) * wb ^ (0.12807 : ℝ))

noncomputable def omegaCb (h Om : ℝ) : ℝ := Om * h ^ 2 - omegaNu

noncomputable def H0 (h : ℝ) : ℝ := 100 * h

noncomputable def DH_over_rd (c h E rd : ℝ) : ℝ := c / H0 h / E / rd

The 10 theorem statements, verbatim (proofs elided):

theorem rd_numerator_pos : 0 < 55.154 * Real.exp (-72.3 * (omegaNu + 0.0006) ^ 2) := ...

theorem rdAubourg_pos {wcb wb : ℝ} (hcb : 0 < wcb) (hb : 0 < wb) :
  0 < rdAubourg wcb wb := ...

theorem rdAubourg_strictAntiOn_cb {wb : ℝ} (hb : 0 < wb) :
  StrictAntiOn (fun wcb => rdAubourg wcb wb) (Set.Ioi (0 : ℝ)) := ...

theorem rdAubourg_strictAntiOn_b {wcb : ℝ} (hcb : 0 < wcb) :
  StrictAntiOn (rdAubourg wcb) (Set.Ioi (0 : ℝ)) := ...

theorem rd_strictAntiOn_Om {h wb : ℝ} (hh : 0 < h) (hb : 0 < wb) :
  StrictAntiOn (fun Om => rdAubourg (omegaCb h Om) wb) {Om : ℝ | 0 < omegaCb h Om} :=
    ...

theorem DH_over_rd_eq_hrd (c h E rd : ℝ) :
  DH_over_rd c h E rd = c / (100 * (h * rd) * E) := ...

theorem DH_over_rd_degenerate {h rd h' rd' : ℝ} (c E : ℝ) (hprod : h * rd = h' * rd') :
  DH_over_rd c h E rd = DH_over_rd c h' E rd' := ...

theorem gate_reparam_exact {hrad : ℝ} (hne : hrad ≠ 0) (c E hrd : ℝ) :
  DH_over_rd c hrad E (hrd / hrad) = c / (100 * hrd * E) := ...

```

```

theorem hrd_mono_generic {a w Om h1 h2 : ℝ} (ha : 0 < a) (ha3 : a ≤ 1 / 3) (hw : 0 < w)
  (hOm : 0 < Om) (hh1 : 0 < h1) (h12 : h1 < h2) (hsupp : w ≤ (1 - 2 * a) * (Om * h1 ^
    2)) :
  h1 * (Om * h2 ^ 2 - w) ^ a < h2 * (Om * h1 ^ 2 - w) ^ a := ...

theorem hrd_strictMonoOn_h {Om wb : ℝ} (hOm : 0 < Om) (hb : 0 < wb) :
  StrictMonoOn (fun h => h * rdAubourg (omegaCb h Om) wb)
  {h : ℝ | 0 < h ∧ omegaNu ≤ (1 - 2 * 0.25351) * (Om * h ^ 2)} := ...

```

In words, the theorems say the following.

- Eq. (1) is positive, and strictly decreasing in ω_{cb} and in ω_b , on $(0, \infty)$.
- At fixed h and ω_b , r_d is strictly decreasing in Ω_m on the sampled support $\{\Omega_m h^2 - \omega_\nu > 0\}$.
- At fixed E , the coded D_H/r_d equals $c/[100(hr_d)E]$.
- The gate’s substitution $r_d \mapsto hr_d/h_{\text{rad}}$ at fixed $h_{\text{rad}} \neq 0$ is an exact reparametrisation.
- Identifiability (secondary r_d): at fixed $\Omega_m > 0$ and $\omega_b > 0$, $h \mapsto hr_d(\Omega_m h^2 - \omega_\nu, \omega_b)$ is strictly increasing on $\{h > 0 : \omega_\nu \leq (1 - 2a)\Omega_m h^2\}$ with $a = 0.25351$. Once BAO fix (Ω_m, hr_d) and BBN fixes ω_b , at most one h on that set matches. The proof uses Bernoulli’s inequality with exponent $a/(1 - 2a) \leq 1$.

The identifiability hypothesis is stronger than the preregistered bare “ $1 - 2a > 0$ ”, because $\omega_\nu > 0$ is subtracted inside the power. It is needed: for eq. (1), hr_d is strictly decreasing in h below $\omega_m = 0.0013$. It holds wherever $\Omega_m h^2 \geq 0.0013$, which includes the T1 posterior by a wide margin ($\omega_m \approx 0.14$). It does not hold on the whole secondary prior support: the band $\omega_\nu < \Omega_m h^2 < 0.0013$ (e.g. $\Omega_m \approx 0.01$ with H_0 of 25–36) is sampled, and there hr_d decreases. The CAMB table’s $\omega_{cdm} \geq 0.005$ support excludes that corner for the primary. As a numerical companion, $hr_d(h)$ is strictly increasing on all 300 sampled curves per box, in the T1 $\pm 5\sigma$ box and in a wide box, for both the eq. 16 and the CAMB-table r_d (fixround_diagnostics.json). The Lean theorem covers eq. 16 only.

The `#print axioms` lines are part of the source file, so the check is reproducible from the source (the Elenchus NO_FOOTPRINT rule). The fix round re-ran the compile read-only against the main checkout’s build. Its first attempt failed with five errors (logged). The second attempt compiled. Its raw output:

```

# started_utc: 2026-09-27T19:01:59Z
# source_sha256: 2b6530838efa0df9bfb475a5d8b25ee29aa121989cc83d04e07f595f5f616e38
'ANSE.BAOBBNH0.rd_numerator_pos'
  depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BAOBBNH0.rdAubourg_pos'
  depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BAOBBNH0.rdAubourg_strictAntiOn_cb'
  depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BAOBBNH0.rdAubourg_strictAntiOn_b'
  depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BAOBBNH0.rd_strictAntiOn_Om'
  depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BAOBBNH0.DH_over_rd_eq_hrd'

```

```

    depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BAOBBNH0.DH_over_rd_degenerate'
    depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BAOBBNH0.gate_reparam_exact'
    depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BAOBBNH0.hrd_mono_generic'
    depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BAOBBNH0.hrd_strictMono0n_h'
    depends on axioms: [propext, Classical.choice, Quot.sound]
rc=0

```

There is no `sorryAx`, and every footprint is the trusted set. `SOCRATEAI-SCIENTIFIC-ELENCHUS elenchus_check.py` reported no findings. A copy with the identifiability proof replaced by `sorry` also compiled with exit code 0 but showed `sorryAx` in its footprint. So the axiom-set rule, not the exit code, is the gate.

Scope. The theorems say nothing about the CAMB r_d of the primary analysis, and nothing about any fitted number. E enters as a free real. In the BBN fits, $E(z)$ depends on h through the radiation term. The “only through hr_d ” identity is therefore exact at fixed E : for the gate, where E is evaluated at fixed h_{rad} . For the full BBN likelihood it holds only up to that radiation dependence, of order 10^{-4} . The Lean docstrings and ledger glosses now say “at fixed E ”. The match between the Lean definitions and the Python code is a reading, filed as a Tier C claim below, not a kernel fact.

Content and deviation from the Lean plan. Six theorems are substantive. They are positivity of eq. (1), its strict monotonicity in ω_{cb} , ω_b and Ω_m , and `hrd_mono_generic` with `hrd_strictMono0n_h`, the preregistered “ $hr_d(h)$ strictly increasing in h ”. An earlier draft lacked the last one. It was added in the fix round, with the strengthened hypothesis described above. The other four theorems are `rd_numerator_pos` (positivity of a numeric constant) and three field identities in which E is a free real (`DH_over_rd_eq_hrd`, `DH_over_rd_degenerate`, `gate_reparam_exact`).

8 Claim ledger

All quantitative claims are registered in an Elenchus tier-capped ledger (`results/bao_bbn_h0/ledger/`): 43 claims, each with a content-addressed evidence blob. The tiers:

- **A:** the 10 Lean theorems;
- **B:** fits, controls and instrument checks;
- **L:** published values, and every comparison with them, including the PASS/PARTIAL verdicts;
- **C:** the Lean-to-code correspondence.

The gate (`ledger.py --evidence-dir`) reports 0 block findings and 2 `LEDGER_UNAUDITED_TIER_A` flags, with exit code 1. The gate returns a non-zero code whenever any finding exists, flags included, so the non-zero exit comes only from those flags. The eight Tier A rows that existed at review carry an audit object from the fix-round *model* referee, which judged every statement faithful. The

object’s `by` field says so, and the evidence is `ledger/referee_statement_audit_fixround.json`. Elenchus defines an audit as a person’s certification, so these rows are model-audited only. No person has audited any statement. The two identifiability theorems were added after that review. Their rows keep `audit: null`, and those are the open flags. An older model review, `ledger/referee_statement_audit.json`, is not used.

Three hand-built mutations were each caught:

- the headline verdict refiled as Tier B (tier inversion);
- a one-byte change to an evidence blob (evidence mismatch);
- a citation refiled as Tier A (kind overclaim).

9 Limitations

- **A reproduction, not a new result.** We use DESI’s data vector, BBN prior and flat priors, so agreement is expected if both pipelines are correct. It tests our pipeline, not cosmology.
- **The DR1 second check is PARTIAL.** A propagation of the DR1-quoted hr_d makes a DESI-side rounding difference plausible, but this was not tested.
- **Model dependence.** The result assumes flat Λ CDM, $N_{\text{eff}} = 3.044$ and standard recombination. The $N_{\text{eff}} = 4.044$ demonstration shifts H_0 by $+4.65$.
- **Instrument details.**
 - The primary r_d is interpolated from a CAMB table, not computed per likelihood call.
 - 22 table nodes far from the posterior were refilled at a neighbouring h .
 - The table’s support truncates the flat H_0 prior at 99. This is relevant only for N3.
 - G1 and G2 evaluate radiation at a fixed h .
- **Eq. 16 accuracy.** The measured deviation of eq. 16 from CAMB (0.029%) exceeds its stated 0.021% at $N_{\text{eff}} = 3.044$. Part of this is the uncorrected N_{eff} offset. The effect on H_0 is the measured -0.034 .
- **Repository checks** (re-run in the fix round; raw logs in `results/bao_bbn_h0/`).
 - `test_rigor_guard.py`: all 141 files pass the anti-stub check.
 - `antigravity_guard.py`: FAIL, with 74 “hallucinated import” flags repository-wide, 0 of them in `scripts/bao_bbn_h0/`. The failure comes from files outside this problem.
 - `pytest tests/` (worktree `.venv`): collection was interrupted by 21 errors, all `ModuleNotFoundError` for `fastapi`, `peft`, `radon`, `starlette` (the worktree `venv` lacks the optional extras), so no test ran. No test covers `scripts/bao_bbn_h0/`, so these results say nothing about this analysis either way.
- **No human review.** No person has reviewed this paper or audited the Lean statements. Eight Tier A rows carry a model referee’s audit, and two carry none.

Reproducibility

Scripts live in `scripts/bao_bbn_h0/`:

- `rd_camb_table.py`, `controls.py`, `fit_real.py` and `make_figure.py` produce the fit;
- `lean_recheck.py`, `write_lean_verification.py`, `build_ledger.py` and `ledger_gate_check.py` produce the ledger;
- `fixround_diagnostics.py` and `control_n4.py` produce the fix-round diagnostics and the post-hoc control N4;
- `make_paper_inputs.py` generates every number in this paper.

The `fit.json` used here has sha256 prefix `8974d5d29ad7bb5b`. MCMC chains are stored off-repository, and their sha256 are listed in `results/bao_bbn_h0/chains_pointer.json`. Software: CAMB 2.0.4, classy 3.3.4, emcee, astropy, numpy and scipy.

References

- [1] DESI Collaboration, arXiv:2503.14738 (2025); eqs. (2), (14), (17), (19), Table V.
- [2] DESI Collaboration, arXiv:2404.03002 (2024); eqs. (2.8), (4.4), Tables 2–3, Sec. 8.
- [3] Aubourg et al., arXiv:1411.1074 (2015); eqs. 16–17.
- [4] N. Schöneberg, arXiv:2401.15054 (2024); eq. (5.1).
- [5] N. Schöneberg et al., arXiv:2209.14330 (2022); Sec. 2.1.1.
- [6] Planck Collaboration, arXiv:1807.06209 (2018); Tables 1–2.